What Can Mongolian Tell Us about Reflexivity?

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Jun 27, 2026

Abstract

Mongolian reflexive binding relies on a dedicated suffix (*-AA*) to mark local coreference between a subject and an object, adpossessor, or embedded subject in a local domain. This suffix strictly obeys Condition A of the Classical Binding Theory. Notably, bound items in reflexive environments must be located at the edge of a leftmost c-commanded phase. The Mongolian reflexive strategies expressed by this suffix cannot be captured by current reflexivity theories, particularly Reuland's (2011, 2018) chain-based analysis, and thus call for a novel approach. These strategies yield the following key insights. **First**, reflexivity is not restricted to coargumental coreference; rather, it extends to non-coargumental coreference relations holding between subjects and adpossessors or between matrix and embedded subjects. It arises as a Spec-Spec dependency constrained by a locality condition based on phase edges, which uniformly explains subject-object, subject-possessor, and subject-subject coreferential relations. Consequently, reflexivity should be defined in terms of syntactic locality rather than thematic-argument structure. **Second**, coreference is a gradient phenomenon. It is regulated as reflexivity when constrained by structural principles such as the Phase-edge Non-intervention Condition (PNC). Governed by PNC, the three reflexivity types constitute a structural continuum. Similarly, reflexivizers, ranging from anaphors to clitics and verbal affixes, form a morphological continuum. These two dimensions merge into a larger macro-continuum, where the blurring of the possessor-possessum relation imparts an intermediate status to subject-possessor reflexivity, effectively bridging the gap to subject-object reflexivity, and the external-argument nature of possessors bridges the gap to subject-subject reflexivity.

Keywords reflexivity, (non)-coargument, possessive, anaphor, Mongolian

Contents

1. Current reflexivity theories: puzzles and difficulties

Reflexivity has often been viewed as a coreferential dependency between two arguments of the same predicate. Listed below are specific claims that align with this central definition.

(1) Reflexivity as coargumental coreference:
  a. A predicate is reflexive iff two of its arguments are coindexed. (Reinhart & Reuland 1993: 663)
  b. Reflexives express an identity relation between corguments. (Everaert 2012: 194)
  c. Reflexivity is a type of interpretation wherein two arguments of the same predicate co-refer. (Cohen & Zribi-Hertz 2014).
  d. A predicate is reflexive iff one semantic argument bears two of the predicate's semantic roles. (Volkova & Reuland 2014: 591)
  e. A reflexive construction is a grammatical construction that can only be used when two argument positions of a clause require coreference and that contains a special form (a reflexivizer) that signals this coreference. (Haspelmath 2023: 20)

This coargument-based definition clearly excludes any configuration from qualifying as reflexive where the semantically identical argument is assigned roles from two different predicates (Volkova & Reuland 2014: 591). Configurations that do not qualify as reflexive but are anaphorically marked often include those in which subject-adpossessor (SP) or subject-subject (SS) coreference is obtained, as exemplified by the Mongolian data in (2b–2c).[1] In (2), two forms, namely the anaphor *öör* 'self' and the reflexive possessive suffix *-AA* (*-aa*, *-öö*, *-ee*, *-oo*), are invariably used for SP, SS, and SO coreference. Beyond Mongolian, many other well-studied languages use a single uniform form (e.g., Japanese *zibun* and Chinese *ziji*) to mark all three coreference types.

(2) a. SO coreference:
  Baatar$_i$ öör$_{i/*j}$-ig-öö šüümžil-sen.
  Baatar self-ACC-RX criticize-PST
  'Baatar criticized himself.' (Bao & Bai 2025: 262–263)

  b. SP coreference:
  Baatar$_i$ öör$_{i/*j}$-in nom-ø-oo mart-san.
  Baatar self-GEN book-ACC-RX forget-PST
  'Baatar forgot his own book.' (Bao & Bai 2025: 269)

  c. SS coreference:
  Baatar$_i$ öör$_{i/*j}$-in buruud-san-ø-aa meder-sen.

[1] The Mongolian data in this paper primarily consist of examples constructed using fieldwork data from informants, who are mostly speakers of the Khorchin dialect in eastern Inner Mongolia (in northern China), and those cited from Guntsetseg (2012), Bai & Cao (2024), Bao & Bai (2025), Bai et al. (2025), and Bai (2026). While the arguments and conclusions presented in this paper are primarily based on data from the Khorchin dialect, they certainly represent the core features of the reflexive strategies of the overall Mongolian language; as a core grammar feature, reflexive strategies essentially do not vary among dialects.

Baatar self-GEN go wrong-PST-ACC-RX admit-PST
'Baatar admitted that he (=Baatar) was wrong.' (Bao & Bai 2025: 265)

A puzzle thus arises: why is reflexivity defined as coargumental coreference, yet the same anaphoric form licenses both reflexivity and non-coargumental coreference cross-linguistically? Let us call this the Uniformity(-of-Coreference-Licenser) Puzzle. This Uniformity Puzzle follows from the claim that anaphors universally license reflexivity but do not enforce it (Reinhart & Reuland 1993; Reuland 2011, 2018; Schadler 2014).

(3) Reflexivity is universally licensed:
   a. A universal property of natural language seems to be that reflexivity must be licensed. A predicate (formed of N, V, etc.) can be reflexive only if it is linguistically marked as reflexive. (Reinhart & Reuland 1993: 662)
   b. Reflexivity must be licensed whenever the inability to distinguish indistinguishables (IDI) comes into play. (Reuland 2018: 97)
   c. A reflexive construction is a grammatical construction that can only be used when two argument positions of a clause require coreference and that contains a special form (a reflexivizer) that signals this coreference. (Haspelmath 2023: 20)

A technical assumption that leads to the Uniformity Puzzle is that the presence or absence of a licenser (a SELF/BODY item) is determined and constrained by the interaction between a syntactic A-chain (Agree-chain) (4) and the internal morphosyntactic structure of anaphors, which may or may not result in reflexivity (Reuland 2011, 2017, 2018; Volkova & Reuland 2014).

(4) a. General condition on Agree-chains (Reuland 2011: 116):
   A maximal Agree-chain (a_1, ..., a_n) contains exactly one link — a_1 — which is both +R and marked for structural Case.
   b. The +R property (Reuland 2011: 116):
   An NP is +R iff it carries a full specification for phi-features.

Below, we briefly outline the core idea of this chain-based analysis using English as an example. In English, SO and SS coreference (in the ECM construction) are marked while SP coreference is not (5).

(5) a. SO coreference:
   Max admires himself.
   b. SP coreference:
   Max$_i$ loves his$_{i/j}$ cat.
   c. SS coreference:
   Max expects himself to pass the exam.

Structurally, *himself* is a DP rather than an NP because it is phi-independent and referential (+R) (Volkova & Reuland 2014; Reuland 2018).

(6) [$_{DP}$ him^{+R} [$_{NP}$ self]

In (5a), because *him* (in addition to the subject *Max*) is +R, *himself* does not satisfy the "contain-exactly-one-link" requirement (4a), leading to illicit chain formation involving two coreferential +R variables on the grid of the verb that are linked to *Max* and *him*. The variables are thus indistinguishable if there is no formal distinction, leading to the "Inability to Distinguish Indistinguishable" (IDI) effect — in cases where a linguistic operation creates indistinguishable tokens of an element in an environment that subsequently leads to their being collapsed, whereas another component of the computational system needs to make a distinction, the derivation fails (Reuland & Everaert 2026: 49). *Self* must therefore be present to distinguish the indistinguishables at the Conceptual-Intentional (C-I) interface. In (5b), SP coreference remains unmarked. *Self* needs not be present because the DP layer, signaled by *his*, acts as a barrier to chain formation that prevents the matrix *v* (or T) head from probing into the nominal domain. IDI thus does not operate. Due to this failed chain formation, SP coreference requires no syntactic marking and exists solely at the interpretive level. In (5c), an illegal chain is formed, and the IDI effect need to be avoided by the presence of *self*. Thus, SS coreference is marked by *himself*; however, it does not constitute reflexivity under (1), because *himself* is not a an argument of the matrix verb. *Self* thus does not licenses reflexivity in (5c). Instead, it is present solely to avoid the IDI effect triggered by the chain. Note that while *himself* is structured as a DP here, its DP layer does not act as a barrier to chain formation where ECM subjects are assigned structural Case by the matrix *v* probe.

Below, we examine the Mongolian anaphor *öör* under (4), in comparison with English *himself*. *Öör* differs from *himself* in both internal structure and clausal behavior. *Öör* is non-referential (–R) as it is phi-deficient (§2.1) and therefore functions as a type of SE anaphor. Categorially, it is analyzed as NP under the chain-based analysis (7).

(7) [$_{NP}$ öör^{-R}]

In (2a), *öör* satisfies (4a) and lacks of a DP layer. Therefore, a legal A-chain is successfully formed and IDI does not operate (or it is safely avoided). *Öör* thus licenses SO coreference as reflexivity, which is a byproduct of the chain formed. In (2b), *öör* functions as a possessor that is coreferential with the subject. Lacking a DP layer to act as a barrier, nothing precludes the formation of a local chain. Once this chain is formed, SP coreference is established and marked by *öör*. However, this chain does not give rise to reflexivity, as the resulting SP coreference fails to satisfy (1). In (2c), *öör* is an embedded subject, which occupies the specifier of the DP representing the structure of the nominalized clause and therefore is local to the matrix *v*. An A-chain is thus formed. However, this SS coreference fails to qualify as reflexivity under (1). In short, under (4), the chain formed is legal as its tail (*öör*) is –R and therefore IDI does not operate for all three types of coreference in (2). However, under (1), only SO coreference is licensed as reflexivity despite the same marking.

Summing up in Table 1, under the chain-based analysis, an Agree-chain, whether it is legal or illegal, can be formed whenever there is no barrier, namely, a phase boundary such as a DP or CP that can insulate the anaphor from the matrix *v* probe, and fails whenever a barrier is present. When a legal chain is formed, a −R item (i.e., a SE anaphor) is present to mark a given type of coreference. In a configuration that allows the formation of an illegal chain, a +R item (i.e., a pronominal with phi-features) must be protected by a SELF/BODY item from being probed by *v* within a local domain.

Table 1. Correlation between Agree-chain and the structure of anaphors (SE vs. SELF)

	Mongolian (NP language)[2]	**English (DP language)**
Structure of anaphor	[$_{NP}$ SE^{-R}]	[$_{DP}$ PRN^{+R} [$_{NP}$ SELF]]
SO coreference (refleixivty)	marked by SE, where a legal chain is formed;	marked by PRN+SELF, where an illegal chain is formed; SELF acts as a *protector*;
SP coreference (not refleixivty)	marked by SE, where a legal chain is formed;	unmarked, where a chain is blocked;
SS coreference (not refleixivty)	marked by SE, where a legal chain is formed;	marked by PRN+SELF, where an illegal is formed; SELF acts as a *protector*;

Obviously, the intertwining of A-chain formation with the categorial status of anaphors and possessive phrases is crucial to Reuland's theory of cross-linguistic variations in coreference types. However, while this chain-based analysis captures a wide range of cross-linguistic facts, it remains inadequate, failing to resolve the Uniformity Puzzle, for the following reasons.

Given that phase boundaries such as DP and CP/vP layers are analyzed as shields against chain formation, any syntactic operation attempting to cross these layers should be blocked in accordance with the Phase Impenetrability Condition (PIC) (Chomsky 2001). Consequently, a SELF form should not need to occur within a DP anaphor simply to prevent or license an A-chain that is entirely external to that DP. Thus, strictly speaking, a SELF form does not occur to protect a pronominal element from an external operation (such as an Agree-chain formed across DP and vP phases); rather, its distribution must be driven by other factors that satisfy the requirements of IDI.

Additionally, a logical leakage exists when correlating Agree-chain formation with the internal morphosyntax of anaphors. To clarify this issue, we can employ the term "phase-as-shield constraint" to denote a situation in which phase categories (DP, vP, and CP) simultaneously shield phase-internal items and act as barriers to phase-external operations. The chain-based analysis is caught in a contradiction: it attributes the *absence* of SELF in SP coreference to the satisfaction of a DP-as-shield constraint, yet it attributes the *presence* of SELF in SO and SS coreference to a vP-as-shield constraint. Alternatively stated, the *presence* of SELF would be treated as a violation of the DP-as-shield constraint in SP configurations (8), but as a requirement in SO/SS configurations (9), despite the uniform phasehood of the DP layer across all three coreference types.

(8) SELF is disallowed because of its violation of a DP-as-shield condition:
[$_{\mathbf{vP=phase}}$ Max admires [$_{\mathbf{DP=phase}}$ his/*himself's [$_{NP}$ boss]]]

[2] Under Reuland (2011) and Despić (2015), Mongolian is an NP language. However, this characterization is untenable. See Section 3.2 (and the sources cited therein) for an argument that Mongolian is a DP language.

(9) SELF is allowed because of its satisfaction of a vP-as-shield condition:

a. [$_{\mathbf{vP=phase}}$ Max admires [$_{\text{DP=phase}}$ him [$_{\text{NP}}$ self]]]

b. [$_{\mathbf{vP=phase}}$ Max expects [$_{\text{TP}\neq\text{phase}}$ [$_{\text{DP=phase}}$ himself] to [$_{\text{vP=phase}}$ pass the exam]]]

Thus, the question arises: why can the DP layer act as a shield in (8) while it cannot in (9)? In a nutshell, Reuland (2011, 2018) and Reuland & Zubkov (2022) handle this asymmetry between SP and SO/SS coreference by assuming that the probing by *v* can pierce the DP layer in (9) but not in (8). Unfortunately, this assumption fails to provide a compelling explanation for the structural uniformity observed between anaphoric object DPs (9) and regular possessive DPs (8), given syntactic constraints like the PIC. The precise reason for this analytical failure is as follows. Note that SELF anaphors can be modified by an adjective, as in *his/their damn self/selves* (Ahn & Kalin 2018). This indicates that anaphoric objects and regular possessive DPs do not structurally differ. The fact that a modifying adjective can be present (e.g., **him/them damn self/selves*) demonstrates that SELF is projected as a noun within the DP, without being syntactically linked to an external operation such as A-chain. Furthermore, the pronominal is genitive-marked, rather than accusative-marked, when an adjective is present, indicating that structural Case assignment by the matrix *v* does not play a crucial role in triggering an A-chain across the external and internal domains of the DP phase, regardless of whether the DP contains SELF or not. Additionally, assuming that the internal DP structure of a SELF anaphor, and hence the presence of *self*, is not finalized until an external head (*v*) probes into it inevitably introduces a look-ahead problem (Bruening 2018).

Thus, under the chain-based analysis that posits distinct operations, the question of why anaphoric objects and possessive DPs under a reflexive interpretation must involve different syntactic operations, which we may call the Possessive Puzzle, remains a significant challenge. This Possessive Puzzle concerns the syntax of different types of coreference while the Uniformity Puzzle concerns their morphology, which mirrors their syntax.

Summing up the problem with the chain-based analysis, while correlating the presence of a SELF/BODY form it with IDI remains conceptually sound, embedding that form within an anaphoric DP while conditioning its presence on an external operation (an Agree-chain) to avoid the IDI effect inherently constitutes a violation of constraints such as the DP-as-shield constraint and PIC (Chomsky 2001) as well.

Beyond these puzzles and problems, the chain-based analysis faces even more critical challenges, particularly its failure to account for cross-linguistic facts from languages like Mongolian. In Mongolian, the same SELF element (*öör*) is contained within object anaphors (10a), regular anaphoric possessive DPs (10b),[3] and anaphoric genitive subjects of nominalized clauses (10c), and it can be optionally omitted in all three environments. Additionally, the reflexive marker (RX) must be present to consistently mark all three coreference types (SO, SP, and SS), regardless of whether *öör* is phonologically realized. This distribution suggests a uniform underlying mechanism wherein the uniform marking (by either *öör* or RX), the optionality of *öör*, and the obligatory presence of RX are all subject to the same locality-sensitive operation. Consequently, the operations driving SO (coargumental) and SP/SS (non-coargumental) coreference do not differ in Mongolian, directly contradicting the predictions of

[3] See Section 3.2 for detailed discussion on the DP status of anaphoric subjects in Mongolian.

Volkova & Reuland (2014: 591) and Reuland (2017: 26; 2018: 95).

(10) a. SO coreference:

Baatar$_i$ (öör$_{i/*j}$-in) bey-ø-ee šüümžil-sen.
Baatar self-GEN body-ACC-RX criticize-PST
'Baatar criticized himself.' (Bai 2026: 13)

b. SP coreference:

Baatar$_i$ (öör$_{i/*j}$-in) nom-ø-oo mart-san.
Baatar self-GEN book-ACC-RX forget-PST
'Baatar forgot his own book.' (Bao & Bai 2025: 268–269)

c. SS coreference (Type I):

Baatar$_i$ (öör$_{i/*j}$-in) buruud-san-ø-aa meder-sen.[4]
Baatar self-GEN go wrong-PST-ACC-RX admit-PST
'Baatar admitted that he (=Baatar) was wrong.' (Bao & Bai 2025: 265)

d. SS coreference (Type II):

Baatar$_i$ öör$_{i/*j}$-ig-öö buruud-san gež med-sen.
Baatar self-ACC-RX go wrong-PST COMP know-PST
'Baatar realized that he (=Baatar) was wrong.' (Bai 2026: 5)

Table 2. Uniformity of coreference-marking across nominals and clauses

Construction	object DP (10a)	possessive DP (10b)	embedded clause	
			Type I (10c)	Type II (10d)
Category	DP		DP containing clause	CP
Bound item	possessor		subject	
Case marking	genitive			accusative
Coreference marking	reflexive suffix (RX)			

More importantly, the fact that the anaphor can be phonologically null in both coargumental and non-coargumental coreference environments provides a strong counterexample to the protection strategy of licensing reflexivity (Reuland 2011, 2018; Reuland & Everaert 2026). Under that account, when a coargumental coreference relation is established, a SELF element must be present to protect the pronominal from being targeted by an external probe.[5] The protection strategy would incorrectly rule out the Mongolian sentences in (10a–10c), in which the SELF item (*öör*) is not obligatory. That is, the protection strategy fails to answer the question in (11):

(11) a. Why does Mongolian allow the absence of a SELF or BODY item (namely *öör* and

[4] However, the presence of the embedded subject in the form of *öör-in* makes the sentence sound a bit less natural to some speakers. When RX is present on *öör-in*, the sentence still sounds less natural than the case with *öör-in* absent but not unacceptable. The lexical semantics of the verb and pragmatics may affect the acceptability.

[5] The absence of the SELF element (*öör*) cannot be accounted for by either Reuland's (2018) object-agreement account or his inherent-Case hypothesis.

bey) while requiring uniform marking (RX) for all three types of coreference?
b. Why does Mongolian disallow a pronominal (+R) within accusative anaphors (§2.1), while allowing two different non-referential (–R) items (namely *öör* and *bey*)?

Also notable is the leftmost-phase requirement on reflexive binding (12) and the phase-in-phase-edge (PIPE) phenomenon (12c–12d) concerning it in Mongolian. These phenomena suggest that what is essential to regulating and licensing reflexivity is phase-based locality rather than the intertwining between a phase-external operation and a phase-internal forms.[6]

(12) a. [$_{XP}$ binder [$_{CP}$ bindee C]]
b. [$_{XP}$ binder [$_{DP}$ bindee D]]
c. [$_{XP}$ binder [$_{CP}$ [$_{DP}$ bindee D] C]]
d. [$_{XP}$ binder [$_{DP}$ [$_{DP}$ bindee D] D]]

The chain-based analysis cannot correctly predict the leftmost-phase requirement and the PIPE phenomenon. This limitation, along with the two correlated puzzles (namely, the Uniformity Puzzle and the Possessive Puzzle) demonstrates the inadequacy of the analysis. A fundamental issue raised by these empirical breakdowns is what exactly is universal about reflexivity and what underlying mechanism regulates it. Specifically, we address the following specific questions:

(13) a. What is the mechanism that regulates and unifies coargumental and non-coargumental coreference as reflexivity?
b. Why is reflexive licensing is non-universal and why do licensing modes vary cross-linguistically among anaphors, clitics, and null forms?
c. What roles do locality and binding play in regulating and licensing reflexivity?

In order to be able to answer these questions, this paper investigates Mongolian (Altaic, spoken in Mongolia and North China), a language that allows a non-anaphoric nominal suffix as a definite reflexivity licenser in both coargumental and non-coargumental contexts.

The remainder of this paper is organized as follows. Section 2 outlines reflexive strategies in Mongolian, demonstrating that the true licenser of reflexivity is the dedicated reflexive suffix *-AA* rather than the anaphor *öör*. The distribution of *-AA* provides empirical evidence that reflexivity can be either coargumental or non-coargumental. Section 3 presents my theoretical analysis, explaining how reflexivity is regulated and constrained within the syntax. I show that the external-argumenthood of possessors and the possessive DP structure of anaphors unify coargumental (subject-object) and non-coargumental (subject-possessor and subject-subject) reflexivity. Furthermore, I argue that binding is an instance of Agreement between phi-features on two specifier elements within a local domain, which yields reflexivity and reveals that reflexivity operates as a continuum rather than a rigid category. Finally, Section 4 concludes the paper.

[6] Exemplifications and justifications of these configurations are presented in Sections 3.1–3.3.

2. Reflexive strategies in Mongolian

Mongolian, which is right in the middle of a theoretical debate regarding the definition of reflexivity, has two types of markers of coreference: the anaphor *öör* and reflexive suffix *-AA*. The anaphor is attributively used before noun phrases, and the suffix is used after them. For a single coreferential relation, they can either cooccur or independently occur within the same nominal projection. However, they greatly differ in their roles in regulating coreference as reflexivity. Below, we outline the basic grammar of them.

Referred to as "өөрий төлөөний нэр (өөрий төлөөний үг) / [Mongolian script] ([Mongolian script])" in Mongolian grammar (Nasunbayar et al. 1982: 303–304; Has-erdeni 1996: 317–318), literally meaning "self pronoun", *öör* is classified into the nominal category as is the case with many other languages. Table 3 summarizes its inflectional paradigm with specification for Case and number.

Table 3. The anaphor *öör* in Mongolian

	Case specification	Cyrillic orthography	Uyghur orthography
öör (singular)	Nominative	өөр (öör)	ᠥᠪᠡᠷ (öber)
	Genitive	өөрийн (öör-in)	ᠥᠪᠡᠷ ᠦᠨ (öber-un)
	Accusative	өөрийг (öör-ig)	ᠥᠪᠡᠷ ᠢ (öber-i)
	Dative-locative	өөрт (öör-t)	ᠥᠪᠡᠷ ᠲᠦ (öber-tü)
	Instrumental	өөрөөр (öör-öör)	ᠥᠪᠡᠷ ᠢᠶᠡᠷ (öber-iyer)
	Ablative	өөрөөс (öör-öös)	ᠥᠪᠡᠷ ᠡᠴᠡ (öber-eče)
	Comittative	өөртэй (öör-tei)	ᠥᠪᠡᠷ ᠲᠡᠢ (öber-tei)
öörsöd (plural)	(Case not indicated)	өөрсд (öörsöd)	ᠥᠪᠡᠷᠰᠡᠳ (öbersed)

As seen in Table 3, *öör* (and its plural form *öörsöd*, not indicated in the table) can be inflected for seven type of Case. The nominative form is null but never occurs as a nominative argument. When used as a subject (of an embedded clause), it must be either accusative- or genitive-marked. When it occurs in a nominative position, it is attached by the reflexive suffix *-öö* to form *öör-öö* (өөрөө / [Mongolian script]), which functions as an adjunct. It has a form specified for number, but not for person and gender. It is not subject to vowel harmony.

RX is known as the marker of "Ерөнхийлөн Хамаатуулах Нөхцөл / [Mongolian script]" in Mongolian grammar (Nasunbayar et al. 1982: 232–233; Daobu 1983: 30–31; Has-erdeni 1996: 426–429), which literally means "universally relating condition/principle/rule".[7] Being subject to vowel harmony, RX has four allophonic variants (*-aa*, *-ee*, *-oo* and *-öö*), which do not differ from each other in their syntactic and semantic functions., RX has four forms (*-aa*, *-ээ*, *-oo*, and *-өө*) aligning with vowel harmony in Cyrillic orthography (Sanzheyev 1973: 81–82; Kullmann and Tserenpil 2015: 108–110) and two ([Mongolian script] and [Mongolian script]) in Uyghur orthography (Nasunbayar et al. 1982: 232; Has-erdeni 1996: 426).

Table 4. The reflexive possessive suffix *-aa* in Mongolian

[7] Descriptive discussions on RX are found in Poppe (1954: 78–84), Anisman (2010: 17–18), Janhunen (2012: 140–141), and Kullmann & Tserenpil (2015: 108–112), in addition to local grammar books. Explanatory discussions are found in studies including Guntsetseg (2012: 101–106), Maki et al. (2015: 67–80), Bai & Cao (2024: 51–52), Gong & Despić (2024, 2026), Bao & Bai (2025), and Bai (2026).

Cyrillic orthography	-аа (-aa)	Used with masculine back-vowel words containing *a* or *u* (e.g., *ah* 'brother' → *ah-aa* 'one's own brother').
	-ээ (-ee)	Used with feminine front-vowel words containing *e*, *ü*, or *i* (e.g., *ger* 'home' → *ger-ee* 'one's own home').
	-оо (-oo)	Used with masculine back-vowel words containing *o* (e.g., *nom* 'book' → *nom-oo* 'one's own book').
	-өө (-öö)	Used with feminine front-vowel words containing *ö* (e.g., *mör* 'shoulder' → *mör-öö* 'one's own shoulder').
Uyghur orthography	ᠪᠠᠨ (ban/ben)	Used with words ending with a vowel (e.g., *aha* 'brother' → *aha-ban* 'one's own brother').
	ᠢᠶᠠᠨ (iyan/iyen)	Used with words ending with a consonant (e.g., *ger* 'home' → *ger-iyen* 'one's own home').

Two points regarding the morpho-phonological properties of RX and relevant glossing in this paper are noted. **First**, when RX is applied to a direct object that would normally take the accusative Case suffix *-ig*, the accusative suffix is mostly unpronounced (*nom-ig* → *nom-ø-oo*) except when it is attached to the anaphor *öör* (*öör-ig* → *öör-ig-öö*). In this paper, I consistently use "-*ø*" to indicate the phonological absence of this accusative marker *-ig*. **Second**, where RX is attached to the genitive anaphor *öör-in*, it requires the consonant *h* to be inserted before it, as in *öör-in-h-öö* (Cinggeltei 1999: 181; Janhunen 2012: 141). The inserted *h* is constantly left out in the examples in this paper. This *h* must not be confused with the minimal noun *h* as in *Ene nom mini-h* 'This book is mine'. The former is epenthetic whereas the latter is not. The latter, which behaves the same way as English *'s* (e.g., *hers* and *John's*), is the exponent of the nominal complement of a possessive D head, as is the case with the body anaphor *bey*. See Kullmann and Tserenpil (2015: 101–102), Bao & Bai (2025), and Bai et al. (2026) for detailed discussion.

2.1. The anaphor *öör*: Anaphoric status

The anaphor *öör* displays syntactic properties as follows.

First, *öör* resists combining with a personal pronominal when used as an object (14).[8] That is, Mongolian disallows the *himself* type of anaphoric forms.

(14) *Baatar ter/tüün öör-ig-öö šüümžil-sen.[9]
Baatar he self-ACC-RX criticize-PST
'Intended meaning: Baatar criticized himself.'

Second, *öör* lacks a specification for person and gender rather than number (15).[10]

(15) Baatar$_i$/biden$_j$ öör$_i$-/öörsed$_j$-ig-öö šüümžil-sen.
Baatar/we self-/selves-ACC-RX criticize-PST
'Baatar/We criticized himself/themselves.'

---

[8] In Mongolian, non-core (oblique) arguments are assigned other Case such as dative, instrumental and ablative. This paper primarily examines accusative objects.

[9] Henceforth, data phrases are presented without indexation, as the coreferential interpretation is unambiguously indicated by RX. Indices are added only where strictly necessary.

[10] In some varieties such Khorchin as well as Khalkha, the singular form *öör* can be used in both singular and plural environments.

Third, *öör* normally requires an antecedent that c-commands it (16a), while allowing exceptions where an antecedent linearly precedes the anaphor (16b). In (16a), *Baatar* is embedded in the subject DP, failing to c-command *öör*, while the subject DP *öör* c-commands it. Therefore, *öör* is coreferential with the subject DP rather than *Baatar* within it. However, in (16b), *öör-in-bey* is coreferential with *Dorž*, which does not c-command but precedes it. Note that *Dorž* is within a clause, which functions as a theme and is hierarchally lower than *öör-in-bey*, which is an oblique argument.

(16) a. Baatar$_i$-in aab$_j$-ni öör$_{*i/j}$-in-bey-ø-ee šüümžil-sen.
Baatar-GEN father-POSS self-GEN-body-ACC-RX criticize-PST
'Baatar's father criticized himself.'

b. Baatar$_i$ Dorž$_j$-in ire-h eseh-ig öör$_{*i/j}$-in-bey-ees-ni asuu-san.
Baatar Dorž-GEN come-INF whether-ACC self-GEN-body-ABL-POSS ask-PST
'Baatar inquired of Dorž whether he (=Dorž) would come.'

Fourth, *öör* can be embedded inside a complex anaphor consisting of it and a body noun (17c). That is, either SELF (*öör*), BODY (*bey*), or SELF'S-BODY (*öör-in-bey*) type anaphors.

(17) a. Baatar öör-ig-öö šüümžil-sen.
Baatar self-ACC-RX criticize-PST
'Baatar criticized himself.' (=(2a))

b. Baatar bey-ϕ-ee šüümžil-sen.
Baatar body-ACC-RX criticize-PST
'Baatar criticized himself.'

c. Baatar öör-in-bey-ϕ-ee šüümžil-sen.
Baatar self-GEN-body-ACC-RX criticize-PST
'Baatar criticized himself.'

Fifth, it is specified for Case and is optionally absent from PF when genitive-marked (18).

(18) a. Baatar (öör-in) nom-ø-oo mart-san.
Baatar self-GEN book-ACC-RX forget-PST
'Baatar forgot his own book.' (=(10b))

b. Baatar (öör-in) buruud-san-ø-aa meder-sen.
Baatar self-GEN go wrong-PST-ACC-RX admit-PST
'Baatar admitted that he (=Baatar) was wrong.' (=(10c))

The possessive pronominal such as *tüün-ne* 'her/his' can occur with the possessive anaphor *öör-in*, where either or both of them can be absent at PF (19).

(19) Baatar (tüün-ne) (öör-in) nom-ø-oo mart-san.[11]
Baatar he-GEN self-GEN book-ACC-RX forget-PST
'Baatar forgot his own book.' (Bao & Bai 2025: 269)

Sixth, it can be bound by either a local or a non-local subject. When locally bound, it must cooccur with RX (20a), and when non-locally bound, it must cooccur with the third-person possessive suffix *-ni*, notated as POSS (20b). That is, the "*öör* … RX" and "*öör* … POSS" patterns are complementary in distribution and function. With either RX or POSS, *öör* is unambiguously coreferential with a subject.

(20) a. Baatar$_i$ Dorž$_j$-ig öör$_{*i/j}$-ig-öö šüümžil-sen gež bod-son.
Baatar Dorž-ACC self-ACC-RX criticize-PST COMP think-PST
'Baatar thought that Dorž criticized himself (=Dorž).' (Bai 2026: 9)

b. Baatar$_i$ Dorž$_j$-ig öör$_{i/*j}$-ig-ni šüümžil-sen gež bod-son.
Baatar Dorž-ACC self-ACC-POSS criticize-PST COMP think-PST
'Baatar thought that Dorž criticized him (=Baatar).' (Bai 2026: 9)

Seventh, it allows sloppy reading in addition to the strict reading, which is marginally available (21).[12]

(21) Baatar öör-ig-öö šüümžil-sen, Dorž bas (teg-sen).
Baatar self-ACC-RX criticize-PST Dorž too do so-PST
'Baatar criticized himself and Dorž did (=Dorž criticized himself/?Baatar), too.'

Eighth, it can be anteceded by a quantified subject (22).

(22) a. Hün$_i$ büri öör$_i$-ig-öö sain gež bod-dog.
person every self-ACC-RX good COMP think-HAB
'Every person$_i$ would think that he$_i$ is good.'

b. Hen$_i$ č bas öör$_i$-ig-öö muu gež bod-dog-güi.
one PRT elf-ACC-RX bad COMP think-HAB-NEG
'No one$_i$ would think that he$_i$ is bad.'

2.2. The reflexive suffix -*AA*: Categorical status and positions

Traditional grammarians refer to -*AA* as a suffix or a particle, without discussing its exact status. Among theoretical studies, Maki et al. (2015: 67ff) labeled RX "pronoun" and Gong & Despić (2024, 2026) treated it as a D head. Maki et al. (2015: 67ff) proposed that RX undergoes LF

[11] None of the variants *tüün-ne öör-in nom-oo* 'his/her own book', *tüün-ne nom-oo*, or *öör-in nom-oo* is more productive or natural than the most compact form, *nom-oo*, in which both the pronominal and the anaphor are omitted. The least compact variant, *tüün-ne öör-in nom-oo*, is the least natural and is even unacceptable to some speakers.

[12] Schadler (2014: 75) also notes that the sloppy reading is preferred to the strict reading.

movement to its antecedent, e.g., the subject. However, RX can never be an element that can be characterized as a reflexive pronoun. It lacks a specification of any nominal features, which a pronoun may have. Functionally, RX is similar to anaphors (or reflexive pronouns), but it should belong to a different category. According to Gong and Despić (2024, 2026), RX as a D head selects *n*P, where *öör*, they assume, is a noun much like a common noun.[13]

In order to describe precise the categorial status of RX, let us first examine its surface position. Linearly, RX occurs in the rightmost position within a phrase requiring it, as in (23–24). Specifically, RX can be attached to either the genitive anaphor *öör-in*, in both possessive DPs (23a) and nominalized clauses (24a) or the accusative-marked head noun (25). When *öör-in* is absent, RX can only be attached to the head noun, following the Case suffix, in both possessive DPs (23b) and nominalized clauses (24b). Note that *gež* clauses do not allow genitive subjects and Case-marking at the clause boundary. Thus, RX must be attached to the accusative subject *öör-ig*, following the accusative marker *-ig* (25).

(23) a. Baatar öör-in-öö nom-ig mart-san.[14]
Baatar self-GEN-RX book-ACC forget-PST
'Baatar forgot his own book.' (adapted from Bao & Bai 2025: 263)

b. Baatar (öör-in) nom-ø-oo mart-san.
Baatar self-GEN book-ACC-RX forget-PST
'Baatar forgot his own book.' (=(2b/10b))

(24) a. Baatar öör-in-öö buruud-san-ig meder-sen.
Baatar self-GEN-RX go wrong-PST-ACC admit-PST
'Baatar admitted that he (=Baatar) was wrong.'

b. Baatar (öör-in) buruud-san-ø-aa meder-sen.
Baatar self-GEN go wrong-PST-ACC-RX admit-PST
'Baatar admitted that he (=Baatar) was wrong.' (=(10c/18c))

(25) Baatar$_{i}$ öör$_{i/*j}$-ig-öö buruud-san gež med-sen.
Baatar self-ACC-RX go wrong-PST COMP know-PST
'Baatar realized that he (=Baatar) was wrong.' (=(10d))

In short, where a genitive subject/possessor occurs, RX comes after either the genitive suffix *-in* or the accusative suffix *-ig* (26a); and where no genitive subject/possessor occurs, RX comes after the accusative suffix *-ig* (26b).

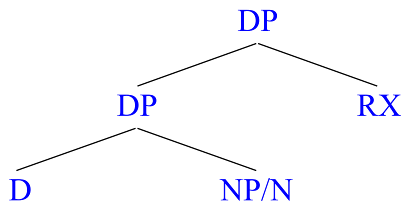
(26) a. [… -gen-RX … -acc …]
[… -gen … -acc-RX …]
b. [… -acc-RX …]

[13] Gong & Despić's (2024, 2026) analysis of RX as a D head is untenable. See Bai (2026) for discussion on why RX cannot be a D head.

[14] The sentence in which RX is attached to *öör-in* (23a, 24a) is less natural and even unacceptable to some speakers.

Below, we discuss the syntactic properties of RX. Since it neither heads nor categorizes a phrase, it is structurally an adjunct in the hierarchy, as represented in (27).

(27) RX ooccurs in the rightmost position within the phrase requiring it: [$_{DP}$ [$_{DP}$]-RX]



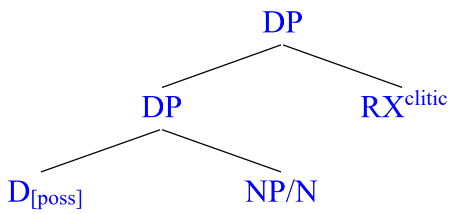
As will be clear in what follows, the syntactic positioning of RX is dependent on a possessive head. Assuming that a genitive DP represents the structural relation of possessors and possessums, RX is attached to either the possessum or the possessor (28). Hierarchically, RX sits in the sister node of the possessive phrase (29a) or the possessor phrase (29b).

(28) a. [$_{DP}$ possessor [$_{NP}$ possessum]-RX]
b. [$_{DP}$ possessor-RX [$_{NP}$ possessum]]

(29) a. b.

Given that RX is an outermost (rightmost) element in a DP, it should occupy an adjunct/specifier position. However, this would imply that it is a phrasal element, contrary to the fact that it does not display any properties displayed by phrases; it lacks a specification for phi-features (person, gender and number) and is unable to move. This in turn indicates that it is a head-like element.[15] This ambivalent property of RX suggests that it is in fact a clitic-like element. Note that clitics are generated as specifiers of a null head (Bošković 1997, 2002). This leads to the following representation (30), in which RX sits in the rightmost specifier of DP.

(30) Position of RX as a clitic[16]

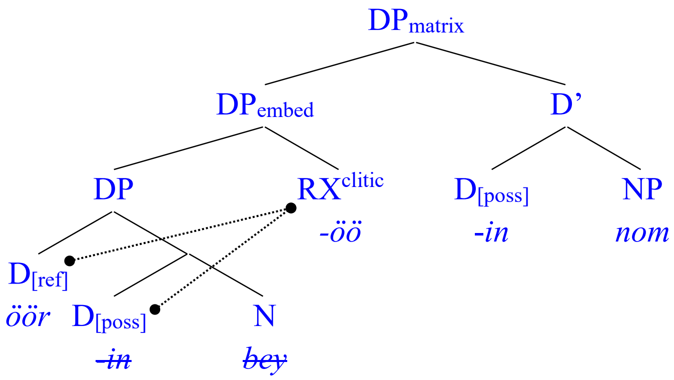


[15] See Chomsky (1995a), Bošković (1997, 2002), and Reinhart (2016) for relevant discussion on the ambiguous property of clitics, which behave like both X^0 or XP.

[16] Notably, RX is a nominal clitic, unlike, for example, what is referred to as the "reflexive clitic" *se* in Romance and Slavic languages, which is identified as a verbal clitic (Reinhart 2016: 189–191).

Given this, RX occurs to the right of NP/N. Note, however, that RX can also occur to the left of N (23a). As shown below, the actual linear string *öör-in-öö nom* 'self's/own book' (31a) may have the structure in (31c), where the spell-out forms of the embedded $D_{[poss]}$ (-*in*) and N (*bey*) will be absent at PF (§3.1), represented by the strikethrough. RX is to the left of the matrix $D_{[poss]}$ (-*in*) in the hierarchical structure (31c). The resulting linear structure would thus like (31b), contrary to the actual linear order (31a). Note that -*in* in the surface structure is the realization of the matrix $D_{[poss]}$, not the embedded $D_{[poss]}$. The embedded $D_{[poss]}$ is absent at PF.

(31) a. Actual linear ordering: *öör*-**GEN-RX** NP (e.g., *öör-**in-öö** nom* 'self's book')[17]
b. Reverse linear ordering: *öör*-**RX-GEN** NP (e.g., *öör-**öö-in** nom* 'self's book')
c. Hierarchical structure ([phi] left out):



Resulting in the non-existent linear string (31b), the hierarchical structure in (31b) might appear incorrect. However, there is solid reasoning for justifying its correctness. The crux of the matter is to determine how RX is constrained by a morpho-phonological principle in addition to syntactic rules. It is evident that RX is not part of an anaphor or a pronoun. It is simply a marker of reflexivity with a status of clitic. Since it is a clitic, RX is not able to project in syntax. Phonologically, RX is subject to vowel harmony, as also suggested by the fact that it has allophonic variants such as -*aa*, -*oo*, -*öö*, and -*ee*. This indicates that what determines the position of RX is not only a syntactic rule such as binding but also a spell-out rule. This said, it is reasonable that the insertion of RX into the structure takes place after the insertion of all other materials required in forming an anaphor and spelling out a possessive relation. This is to say that $D_{[ref]}$ in the embedded DP (DP_{embed}) and the matrix $D_{[poss]}$ are spelled out first by *öör-in*, which is inserted as an ***inseparable morphological unit***, and then RX is inserted into a ***linear, not hierarchical, position adjacent to* öör-in**, resulting in *öör-in-öö*. Let us refer to this as "the Late Insertion Rule", which is informally stated as follows.

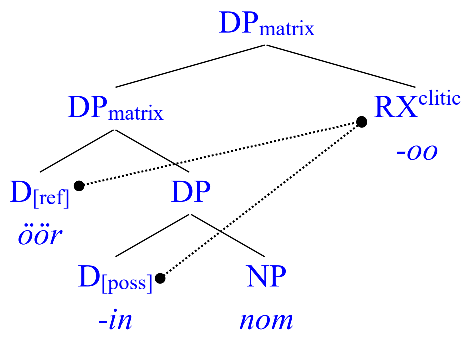
(32) The Late Insertion Rule (LIR)
RX is inserted after the morpheme(s) required for DP formation are inserted.

LIR, essentially a spell-out rule, interacts with a syntactic condition to describe below. RX being

[17] This nested DP structure is also suggested by English *your all's* (Deal 2006), where $D_{[phi,\ poss]}$ (after fusion) and N in the embedded DP are spelled out by *you-r* and *all* respectively, and the matrix $D_{[poss]}$ is spelled out by *'s*.

subject to LIR implies that the right Spec of DP$_{\text{embed}}$ is empty in syntax. Notably, RX can also be inserted into a position to the right of *nom*, specifically the right Spec of DP$_{\text{matrix}}$, forming *öör-in nom-oo*. In that case, *öör-in* can be absent at PF as it is not the host of RX.

(33) a. Actual linear ordering: *öör*-**GEN** NP-**RX** (e.g., *öör-**in** nom-**öö*** 'self's book')
b. Hierarchical structure ([phi] left out):



As regards (31c), the position of RX is within the embedded DP in the syntax because RX is required to be ***hierarchically adjacent to the possessive DP requiring it***. That is, it is required to be projected in the right Spec of DP$_{\text{embed}}$, which requires it, in the syntax. RX is required by DP$_{\text{embed}}$ because there is D$_{\text{[poss]}}$, which heads the embedded possessive structure (DP$_{\text{embed}}$). The common property of (31c) and (33b) is that the syntactic position of RX c-commands the possessive D head (D$_{\text{[poss]}}$), which determines where RX is projected in the syntax. Put another way, ***D$_{[poss]}$ wants RX to be adjacent to its own projection***.[18] Let us refer to this property as "Adjacent Condition", which is informally stated as follows (34).

(34) Adjacent Condition (AC):[19]
RX is hierarchically a sister of DP, D being a possessive determiner head and its Spec bound by the subject of a local inflected clause.

The syntactic position of RX is thus determined by AC and its linear position is determined by LIR, which is preceded by AC but has an adjusting effect on it during the process of derivation. AC accounts for and is supported by the fact that embedded *gež* clauses allow neither a genitive subject nor a rightmost RX (25), whereas embedded clauses without *gež* allow both a genitive subject and a rightmost RX (24). *Gež* clauses are CPs and therefore are not subject to nominalization, unlike object clauses without *gež*, which are TPs (§§3.1–3.2). Failing to have a nominal status, they cannot be selected by D$_{\text{[poss]}}$, and therefore do not satisfy AC. Due to the lack of D$_{\text{[poss]}}$, genitive subjects are disallowed in *gež* clauses. Recall that the accusative subject *öör-ig* in *gež* clauses must not be absent from PF (25). This is because a rightmost RX is disallowed where D$_{\text{[poss]}}$ is lacking, due to AC (specifically due to the lack of D$_{\text{[poss]}}$). In contrast,

[18] An alternative way to interpret this is that RX (the marker of reflexivity) need to be as possible as close to the locus of reflexivity (i.e., D$_{\text{[ref]}}$), whether it is fused with D$_{\text{[poss]}}$ or not, in the syntax.
[19] AC as well as LIR is not presented as a universally holding principle. For the moment, it remains a stipulative formulation of a language-specific property. Where any cross-linguistic facts are found to be supportive of it, AC as well as LIR would prove to be a more general property.

nominalized clauses (structurally, TP dominated by a possessive DP) allow a rightmost RX because they are headed by a possessive D head, which ensures that AC is satisfied by allowing the rightmost RX.

2.3. The reflexive suffix -*AA*: Binding properties and local requirements

While functioning as a reflexivizer, RX differs from anaphors in both Mongolian and other languages and verbal clitics found in, for instance, Romance languages.[20] RX is neither pronominal nor anaphoric since it is underspecified for Case and inflection. Yet, it displays important properties indicating that it manifests Condition A of the Classical Binding Theory (Chomsky 1981).

First, RX is never attached to a nominative argument (35).

(35) *Nom-oo mart-gd-san.
book-RX forget-PS-PST
'Intended meaning: One's own book was forgotten.'

Second, it is licensed by the structural subject of a local clause.[21] In (36), with RX, *darga* 'boss' must denote the boss of Dorž rather than Tuya's.

(36) Tuya Dorž-ig darga-d-aa uurla-san-ig med-sen.
Tuya Dorž-ACC boss-DAT-RX be angry-PST-ACC know-PST
'Tuya knew that Dorž was angry with his (=Dorž's) boss.' (Guntsetseg 2012: 231)

The same is true for the case in which the anaphor *öör* is overt. In (37), *öör* must be coreferential with the local subject *Dorž* rather than the long-distance subject *Baatar*.

(37) Baatar Dorž-ig öör-ig-öö šüümžil-sen gež bod-son.
Baatar Dorž-ACC self-ACC-RX criticize-PST COMP think-PST
'Baatar thought that Dorž criticized himself (=Dorž).' (=(23a))

Third, it is undermined by switch reference. In both (38a) and (38b), the object phrase contains an object relative clause. However, (38a) is grammatical, whereas (38b) is not. This is because the subject of the relative clause is coreferential with the matrix subject in (38a), but not in (38b). That is, (38b) contains switch reference, but (38a) does not.

(38) a. Baatar sur-san hičeel-ϕ-ee mart-san.
Baatar learn-PST lesson-ACC-RX forget-PST
'Baatar forgot the lesson that he (=Baatar) learned.' (=(33a))

b. *Baatar bagš-in zaa-san hičeel-ø-ee mart-san.

[20] Unlike reflexive clitics in Romance languages, RX itself does not fill an argument slot.

[21] Embedded clauses concerned can also be adverbial clauses and depictive predicates. However, they are not discussed in this paper due to space limitation. See Guntsetseg (2012), Maki et al. (2015), Peters (2024), and Gong & Despić (2024, 2026) for relevant discussion.

Baatar teacher-GEN teach-PST lesson-ACC-RX forget-PST
'Intended meaning: Baatar forgot the lesson that his (=Baatar's) teacher taught.'
(Bai & Cao 2024: 51)

The same is true for possessives. Switch reference of the possessor is involved in (39a), but not in (39b), leading to the ungrammaticality of the latter.

(39) a. Tuya baišing-in-aa haalg-ig zas-san.
Tuya house-GEN-RX door-ACC repair-PST
'Tuya repaired the door of her (Tuya's) house.' (Guntsetseg 2012: 102)

b. *Tuya baišing-in haalg-ϕ-aa zas-san.
Tuya house-GEN door-ACC-RX repair-PST
'Intended meaning: Tuya repaired the door of her (Tuya's) house.'

Importantly, these properties resemble the properties of Condition A of the Classic Binding Theory — an anaphor must be bound within its binding domain. Three uncontroversial facts about Condition A are notable in English and many others (40). **First**, the reflexive marker *self* is out in the nominative position (40a). **Second**, anaphors, which contain *self*, are bound by a local subject (40b). **Third**, rebinding is disallowed (40c).

(40) a. *Chris$_i$ said [$_{CP}$ that himself$_i$ was appealing].
b. John made Heidi$_i$ love herself$_i$.
c. [$_{CP}$ Heidi$_i$ believes [$_{DP}$ Martha$_j$'s description of herself$_{*i/j}$]].

This resemblance between Mongolian and English is summarized in (41).

(41) Binding properties of RX, in comparison with *self*:

Mongolian RX	**English *self***
Disallow switch reference	Disallow rebinding
Resist a nominative argument position	
Bound by a local subject	

While the nominative-resisting and local-binding properties are straightforward for both RX and *self*, the comparison between switch reference for RX and rebinding for *self* needs clarification. In a nutshell, they are manifestations of the derivational nature of reflexive binding. For example, in (38a), the lesson is a participant of the teaching event initiated by the teacher in the relative clause, and therefore it is "the teacher's lesson", whether the lesson already has a "possessor". When it participates in another event such as forgetting initiated by a person other than the possessor, it is associated with two different entities, the subject (i.e., Baatar) and the possessor (i.e., the teacher). That is, there are now two relations: possessor-possessum (i.e., teacher-lesson) and subject-object (Baatar-lesson). Both relations involve the lesson while also involving disjoint reference between the teacher and Baatar; that is, the possessor of the lesson is not coreferential with the subject. This leads to the failure of RX. In short, during the derivation, a DP (e.g., *hičeel* 'lesson') cannot be re-associated with a non-coreferential D.

This re-association resembles the rebinding disallowed for Condition A. For example, in (40c), *herself* is bound by *Martha* within DP and is inaccessible for rebinding by *Heidi* within CP.

3. Interpreting and structuring reflexivity

3.1. Anaphors as possessive DPs

In interpreting and structuring reflexivity, we first need to clarify the possessive DP nature of anaphors. The possessive DP nature is evidently indicated by three interchangeable forms of anaphor in Mongolian: SELF (*öör*), BODY (*bey*), and SELF'S BODY (*öör-in bey*), as exemplified in (17), repeated below as (42).

(42) a. Baatar öör-ig-öö šüümžil-sen.[22]
Baatar self-ACC-RX criticize-PST
'Baatar criticized himself.' (=(2a/27a/28a))

b. Baatar bey-ϕ-ee šüümžil-sen.[23]
Baatar body-ACC-RX criticize-PST
'Baatar criticized himself.' (=(28b))

c. Baatar öör-in-bey-ϕ-ee šüümžil-sen.
Baatar self-GEN-body-ACC-RX criticize-PST
'Baatar criticized himself.' (=(28c))

The contrast between (42b) and (42c) shows that *bey* in its anaphoric use is logically a possessum, possessed by the metaphorical possessor *öör*. In the syntax, this dependency is mediated by the possessive determiner head *-in* (43).

(43) $[_{XP}$ … *John*$_{i}$ … $[_{DP}$ *öör*$_{i}$ $[_{D'}$ *-in* $[$*bey*$]]]]$

Analyzing anaphors as possessive DPs is also supported by and accounts for Schladt's (2000) finding that in a worldwide sample of 148 languages, approximately 60.1% (89 out of 148) used constructions that originated from body part terms. Importantly, SELF-anaphors (in English-like languages) are structurally analogous to body anaphors (44a), specifically regarding their status as possessive DPs (Ahn & Kalin 2018; Bai et al. 2026). This is evidenced by (44b), in which an adjective modifies *self* and the pronominal is genitive-marked, suggesting its status as a possessor.

(44) a. $[_{XP}$ … *Heidi*$_{i}$ … $[_{DP}$ *her*$_{i}$ $[_{D'}$ $[$*self*$]]]]$
b. his/their damn/younger/own self/selves

[22] In (42a), *öör* is an object in the surface. However, it is in fact the possessor within a possessive DP *öör-in-bey* 'self's body', where the genitive marker *-in* and the minimal noun *bey* 'body' are absent at PF. See Section 3.1 for detailed discussion on its structural representation.

[23] See also Janhunen (2012: 141) and Bai (2026) on the anaphoric status of *bey*.

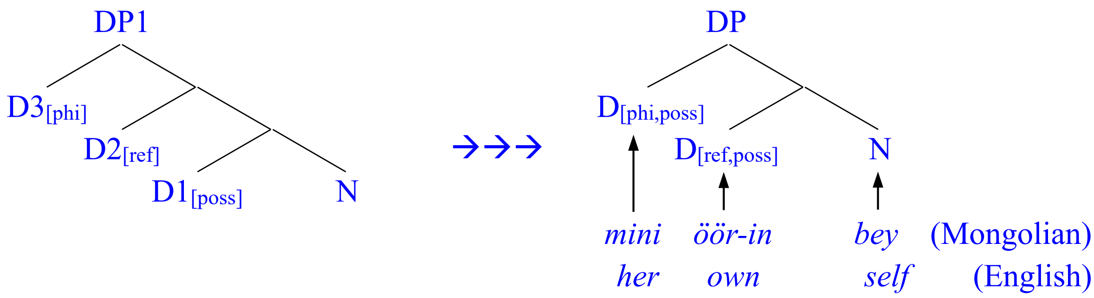
Under a framework like Distributed Morphology (Embick & Marantz 2008), the morphological units making up the anaphor can be analyzed as spelling out one or more single terminal nodes resulting from an operation such as fusion, as significantly demonstrated by Bai et al. (2026). The possessive determiner head D selects a nominal with minimal lexical content, such as *bey* 'body' in Mongolian (45a) or *self* in English (45b).[24]

(45) a. Structure of *öör-in bey* b. Structure of *herself*

Let us elaborate on this analysis. Following Bai et al. (2026), three formal features including possession [poss], reflexivity [ref], and phi [phi] enter the derivation separately as determiner (D) heads, with [poss] being the most prominent. In other words, the head of the entire structure is a determiner with a possession feature [poss], notated as $D1_{[poss]}$, which takes two bare D heads (i.e., $D_{[ref]}$ and $D_{[phi]}$) as its multiple specifiers and selects a noun N as its complement.[25] Afterwards, [poss] undergoes fusion with either of [ref] and [phi], leading to the feature clusters, to which particular morphemes are inserted by the Vocabulary Insertion rule (46b). Specifically, *mini* 'my' and *öör-in* 'self's; own' are inserted to $D_{[phi,poss]}$ and $D_{[ref,poss]}$ in Mongolian, while *her* and *own* are inserted to them in English. The morphemes *bey* 'body' and *self* realize N in each language, respectively.

(46) Derivation of anaphors (as reduced possessive DPs)

a. Initial structure b. Ultimate structure



While the possessive DP structure under discussion is universal, the way of lexicalizing it as an anaphor varies both within and among languages. Given economy principles such as Minimize Exponence (53), it is expected that morphological redundancy will be reduced, such that any of these elements ($D_{[phi,poss]}$, $D_{[ref,poss]}$, and N) can, and often must, be null at PF. For example, English *own*, the spell-out form of $D_{[ref,poss]}$, may be absent, leading to the form *herself*, or

[24] By "minimal noun", I mean that a nominal lacking a full lexical content.

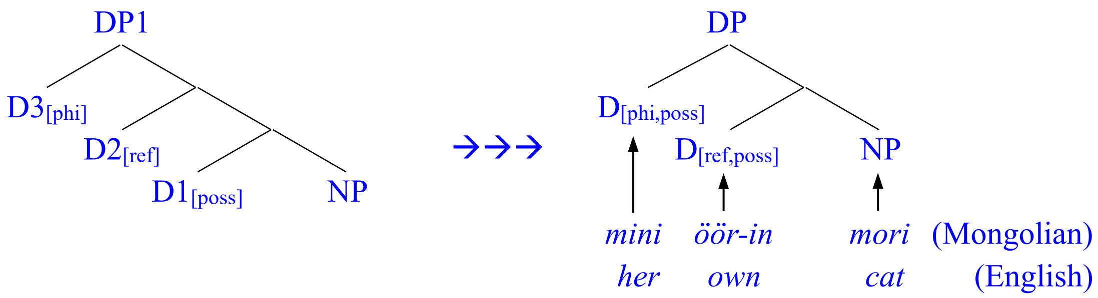
[25] Davis (2023) claims that possessive pronouns are derived from a structure in which the possessive D head selects a non-projecting D head with phi-features; this is effectively a bare phrase under *Bare Phrase Structure* (Chomsky 1995b). Following this, Bai et al. (2026) demonstrate that the head-phrase distinction is collapsed for both possessive pronouns and anaphors.

present, leading to the form *her own self*, where *her* is a metaphorical possessor (Bai et al. 2026). In contrast, in Mongolian, possessive personal pronouns such as *mini*, the spell-out form of $D_{[ref,poss]}$, are preferably or obligatorily null, leading to the form *öör-in bey*.[26] *Öör-in bey* can further be condensed to *öör* or *bey* on economical grounds.

The same applies to regular possessive DPs with reflexive interpretations, where the complement of D is a content noun NP such as *mori* 'horse' and *cat* (47). Similarly, $D_{[phi,\ poss]}$ is null in Mongolian, as is $D_{[ref,\ poss]}$ in English.

(47) Derivation of regular possessive DPs (full possessive DPs): reflexive

a. Initial structure b. Ultimate structure

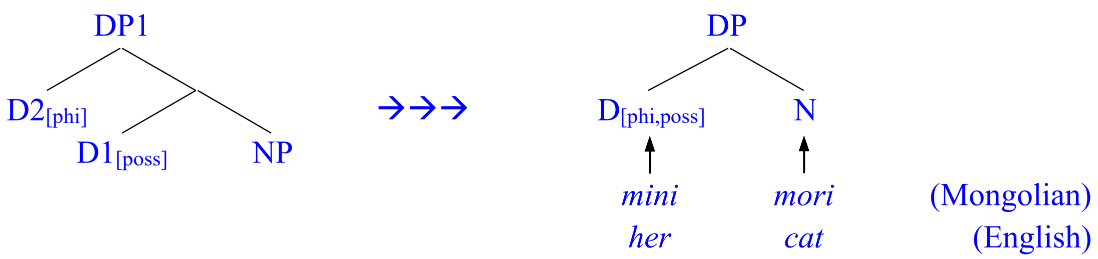


Given that reflexivity is conditioned by binding, it is reasonable to suggest that [ref] is evoked by binding as [phi]-valuation under the Binding-as-Agreement analysis (Reuland 2005, 2011, 2020), wherein [phi] on the antecedent value those on the bindee via agreement within the binding domain. I contend that $D_{[phi]}$ in the specifier of DP, rather than the entire possessive structure, serves as the bindee and that binding as [phi]-valuation evokes [ref]. That is, reflexivity arises wherever binding obtains.

In a structure without binding, [ref] stays inert. This is the case for non-reflexive regular possessive DPs such as *mini mori* 'my horse' in Mongolian and *her cat* in English, where the anaphor *öör* and *own* are not projected in the syntax (48).

(48) Derivation of regular possessive DPs (full possessive DPs): non-reflexive

a. Initial structure b. Ultimate structure



Note that under the proposed analysis, pronominal possessors spell out bare $D_{[phi]}$ heads, which are functionally equivalent to bare phrases and structurally projected as specifiers. In contrast, under the chain-based analysis, they spell out the D head of the entire DP structure.

[26] As far as Mongolian and English are concerned, the reason $D_{[phi,poss]}$ is obligatorily realized in English but not in Mongolian is clear: English requires overt determiners, whereas Mongolian allows null determiners. In English, $D_{[phi,poss]}$ must be realized to satisfy this requirement for an overt D head. See Bai et al. (2026) for more detailed discussion.

An important issue that needs clarification is that Mongolian is a DP language. Under Reuland (2011, 2018) and Despić's (2015) analysis, Mongolian would be classified as an NP language since it does not have over definiteness marking and disallows a pronominal within object anaphors. However, Mongolian proves to be a DP language, as argued by Aravind (2021), Bai et al. (2025), Lim (2025a, 2025b), and Bai (2026). Aravind (2021) challenges Bošković's (2005, 2008) bare NP hypothesis by demonstrating that Mongolian nominalized clauses project a full DP layer acting as a strict syntactic phase. Lim (2025a, 2025b) challenges rigid nominal classifications by suggesting structural, size-variant co-existence rather than a binary DP/NP split in Khalkha Mongolian. Following Bai et al. (2025) and Bai (2026), this paper demonstrates that the status of anaphoric/pronominal DPs relies on their possessive nature, as represented above. That is, the DP status of Mongolian anaphors has little to do with definiteness. Rather, it is a manifestation of their possessive nature. In other words, the DP structure assumed here for Mongolian anaphors does not mean that Mongolian is a definiteness-marking language; rather, it is a possessive-marking language. Crucially, possessive phrases structurally project as phasal DPs. Highly relevant to this distribution is the possibility that two distinct DP layers project above the NP, encoding definiteness and possessivity, respectively.

3.2. Binding as phi-Agreement

The proposed analysis of this decomposed anaphoric structure opens the way for a formal implementation of binding. Following Reuland (2001, 2011), this paper argues that binding is an Agreement operation between phi-features of the binder and the bindee. As shown in (49), [phi] on the antecedent subject and that in Spec (bare D phrase) of the $D_{[poss]}$ head agree, instantiating binding.

(49) Binding as phi-Agreement:

$[_{TP}\ [_{VoiceP}\ \text{sbj}_{[phi]}\ [_{DP} \ldots \text{D}_{[phi,\ poss]} \ldots \text{D}_{[ref,\ poss]} \ldots [_{NP} \ldots] \ldots] \ldots] \ldots]$

The proposed analysis, wherein binding takes place between specifiers, successfully accounts for the binding of accusative and genitive anaphors as well as binding into nominals and inflected clauses, with or without the complementizer *gež*, in a unified way.[27]

(50) a. Complement clause as CP dominating TP:

Baatar	öör-ig-öö	buruud-san	gež	med-sen.
Baatar	self-ACC-RX	go wrong-PST	COMP	know-PST

‘Baatar realized that he (=Baatar) was wrong.’ (=(10d/25))

b. Complement clause as DP dominating TP:[28]

Baatar	(öör-in)	buruud-san-ø-aa	meder-sen.
Baatar	self-GEN	go wrong-PST-ACC-RX	admit-PST

‘Baatar admitted that he (=Baatar) was wrong.’ (=(10c/18c))

[27] See Gong (2022: ch.2) for discussion of the status of *gež* as a complementizer.

[28] Accusative subjects are also possible in TPs (Guntsetseg 2012; Peters 2024), but less acceptable across dialects.

We first need to determine the position of the accusative and genitive subjects. As argued by Fong (2019), Gong (2023), Gong & Despić (2024, 2026), and Bai (2026), accusative subjects of *gež* clauses are raised into Spec of CP, thereby becoming able to be assigned accusative Case by the matrix verb (51). This phenomenon is known as "Hyper-raising" (Fong 2019). Thus, *öör-ig-öö* is within the minimal matrix phase and bound by the matrix subject.

(51) Complement clause as CP dominating TP (50a):

[$_{TP}$ [$_{vP}$ Baatar$_i$ [$_{CP}$ öör$_i$-ig-öö [$_{TP}$ t_i [$_{vP}$ t_i buruud] -san] gež] med] -sen]

To elaborate on binding of hyper-raised subjects in (51), bare D heads with an unvalued [phi] and [ref] is introduced as an external argument by *v*, and subsequently moves to Spec of CP through Spec of TP. Ultimately located in Spec of CP, the D heads are accessible from the matrix *v*. This makes it possible that [phi] is valued by that on the matrix subject via Agree, thereby evoking [ref]. After fusion applies, VI chooses the morpheme *öör* for [ref], with [phi] remaining unrealized.

Hyper-raising does not occur in nominalized clauses in Mongolian (50b). In Mongolian, complement clauses not marked by the complementizer *gež* undergo nominalization (Aravind 2021; Gong 2023; Gong & Despić 2024, 2026; Bai 2026). Categorially, they are DPs, where their subject occupies Spec of DP headed by the possessive D head, which is realized as the genitive marker *-in* (Bai 2026). This is plausible given that TPs are subject to nominalization (Kornfilt & Whitman 2011, 2012). That is, the genitive anaphor *öör-in* as the subject of object clauses such as (50b) is located in Spec of DP that contains a clausal structure. Specifically, TP is selected by D$_{[poss]}$ with the *HOLD* semantics (Bai 2026). In this sense, the genitive subject is some kind of possessor that holds the event as a possessum under its control.

(52) Complement clause as DP dominating TP (50b):[29]

[$_{TP}$ [$_{vP}$ Baatar$_i$ [$_{DP}$ öör$_i$ D(=in) [$_{TP}$ [$_{vP}$ t_i buruud] -san-aa]] meder] -sen]

To elaborate on binding in nominalized clauses, bare D heads with an unvalued [phi] and [ref] are introduced as an external argument by *v*, and subsequently moves to Spec of DP. Located in Spec of DP, the D heads are accessible from the matrix *v*. This makes it possible that [phi] on *öör* is valued by that on the matrix subject via Agree, thereby evoking [ref]. After fusion applies, VI chooses the morpheme *öör-in* for D$_{[ref, poss]}$. D$_{[phi, poss]}$ is realized as the third person singular possessive pronoun *tüün-ne*, which, however, is mostly absent due to economy principles such as (53). Interestingly, such principles even allow the spell-out form of D$_{[ref, poss]}$ (i.e., *öör-in*) to be absent at PF, where RX is available elsewhere, as exemplified by (24).[30]

(53) Minimize Exponence:

The most economical derivation will be the one that maximally realizes all the formal features of the derivation with the fewest morphemes (Siddiqi 2006: 14/82).

[29] A similar structure is assigned to relative clauses. Detailed discussion is available in Bai (2026).
[30] The presence of *öör-in* exerts an intensifying effect on the possessor. That is, *öör-in* is normally null at PF unless an intensification requirement arises.

It is important to note here that locality in cross-clausal binding is achieved by raising the embedded subject to the edge of the maximal CP phase for *gež* clauses and by nominalizing clauses without *gež* into possessive DPs and locating the anaphoric element in their edges. Such cross-clausal binding and within-clausal binding share a common structure mimicking a possessive DP, where reflexive binding is a linkage between specifiers, not simply between a specifier and a complement, in a local domain.

3.3. Reflexivity as Spec-Spec dependency

The fact that the anaphor *öör* is always located in a specifier suggests that reflexivity emerges from binding between two items in specifiers. Additionally, $D_{[phi]}$, occupying the specifier, is the receiver of a phi-value of the antecedent. Regarding English type anaphors/possessives such as *herself* and *her cat*, the possessive pronoun *her* spells out the cluster of the D head and its specifier (i.e., $D_{[phi,\ poss]}$) in a portmanteau fashion. Although it does not directly spell out the specifier of DP alone for morphological reasons, it is somehow transparent morphologically, as evidenced by the semi-regularity observed in *he-r*, *thei-r*, *you-r*, and *ou-r*. The same is true for *hi-s*, *it-s*, and *one-s*.[31] In this sense, the consonants *r* and *s* represent the possessive head $D_{[poss]}$, while the preceding element represents the phi head $D_{[phi]}$ in the specifier of $D_{[poss]}$. Additionally, $D_{[phi]}$, occupying the specifier, is the true receiver of a phi-value; that is, the specifier of DP, rather than the entire DP, acts as the bindee. We may thus refer to this as the "the Bindee-in-Specifier Generalization" (54), which states that bindees are consistently located in a specifier position.

(54) The Bindee-in-Specifier Generalization (the BIS Generalization):

	What is in specifier	**Example**
Body-anaphor	anaphor	*öör-in bey* 'self's body'
***SELF*-anaphor**	personal pronoun	*yourself*
Possessive anaphor	personal pronoun; anaphor	*your (own) car*; *öör-in nom* 'self's book'
Embedded clauses	subject	(50)

Given the BIS Generalization, reflexivity, conditioned by binding, arises from a common underlying structure in (55).

(55) $[_{XP} \ldots \text{binder}_i \ldots [_{YP} \text{bindee}_i\ [_{Y'}\ [N/NP]]]$

Reflexivity can thus be defined as (56), supplementing the definitions in (1).

(56) Reflexivity is a Spec-Spec dependency.

Under the novel definition in (56), the SP coreference as in (57) qualifies as a true reflexivity

[31] See Bai et al. (2026) an explanation of why the accusative forms *himself* and *themselves*, rather than the genitive forms *isself* and *theirselves*, are used in Present-day English.

despite lacking the overt marker *self*. This means that (SP) reflexivity is not universally licensed.

(57) Pasha$_i$ loves her$_i$ cat.

As outlined in Section 1, the coargument-based definition in (1) and licensing requirements in (3) preclude SP coreference as reflexivity simply because it is non-coargumental and there is no overt reflexive marking.[32] However, as strongly evidenced by Mongolian data, SP coreference can be reflexivity universally when it satisfies syntactic locality constraints. Regarding the lack of marking, absence of evidence is not evidence of absence. That is, the fact that anaphors are not present as reflexive markers in such predicates cannot be used as evidence for claiming that reflexivity is not present. The presence or absence of a reflexive marker across languages is due to the parametric properties of individual languages and their morphological inventories of DPs and possessive structures. Consequently, this is not a question of whether reflexivity is present, but rather of whether it is overtly marked.

The definitions in (1) thus hold true only for SO reflexivity. It can explain neither SP reflexivity nor SS reflexivity. In contrast with the narrow definition in (1), the novel definition in (56) aligns more closely with the essential nature of reflexivity.

(58) Reflexivity identification (this paper vs. the literature):

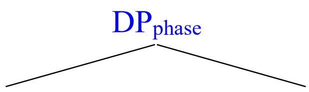
This paper	**The literature**
SO reflexivity	Reflexivity (coargumental reflexivity)
SP reflexivity	Not reflexivity
SS reflexivity	Not reflexivity

A crucial question ((12a), repeated here) arises: what mechanism underlies SO, SP, SS reflexivity? As will be clear in the following discussion, RX is present only when a bound item occurs in the specifier of a leftmost phase in a c-commanded domain. Let us refer to this property as “Leftmost Edge Condition (LEC)” (59).

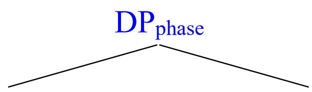
(59) Leftmost Edge Condition (LEC):
Reflexivity arises from binding into a leftmost (or outermost) phase edge.

The mechanism centered around LEC is illustrated below for both SO (coargumental) and SP/SS (non-coargumental) reflexivity. In each structure, phi-features, which agree with those of a c-commanding subject, occupy the edge of the leftmost DP phase, rendering them accessible to a higher probe. Crucially, the maximal phasal projection in (60a) does not block binding into its edge.

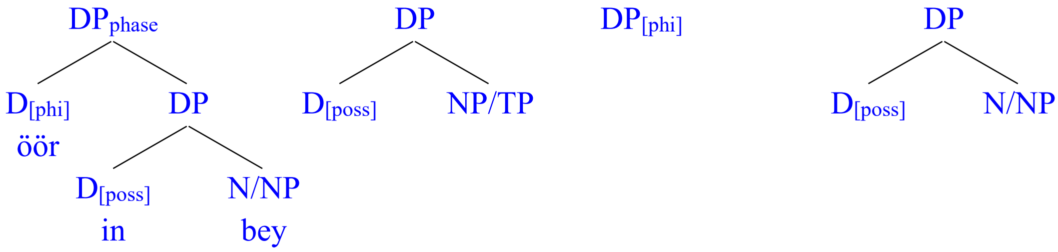
(60) a. For SP/SS reflexivity:



b. For SO reflexivity:



[32] Bai et al. (2026) argue that *her* in such cases is accompanied by a hidden reflexivizer *own*, which is present in the syntax.

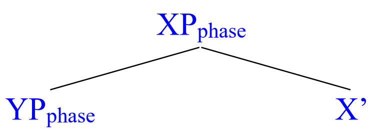


Importantly, the definition based on LEC in (59) provides a blueprint that unifies not only coargumental and non-coargumental coreference as reflexivity, but also regular possessive nominals (including complex anaphors and full possessive DPs) and nominalized clauses. Additionally, LEC successfully captures reflexivity obtained across TP and CP complement boundaries cross-linguistically. In (61), *himself* is base-generated within the embedded TP, which itself is not a phase. *Himself* is outside the embedded vP phase and therefore is accessible from the matrix vP phase which contains the antecedent subject. Crucially, it is *him* ($D_{[phi]}$) in *himself*, rather than *himself*, that is the bindee by virtue of possessing phi-features, satisfying (56) —Binding as phi-Agreement. With *him* being located in Spec of the possessive DP (*himself*) and locally bound, binding with ECM satisfies LEC.[33] Binding of ECM subjects in English and hyper-raised subjects in Mongolian under LEC unifies reflexive dependencies obtained across TP and CP complement boundaries cross-linguistically. What makes the two types of subjects differ is that ECM subjects occupy Spec of TP (61a) while hyper-raised subjects occupy Spec of CP (61b) (§3.2). Unlike ECM and hyper-raising subjects, subjects of nominalized clauses are projected as Spec of DP (61c) (§3.2).

(61) a. ECM (English):
[Max$_i$ expects [$_{TP}$ himself$_i$ to [$_{vP(phase)}$ pass the exam]]]
b. Hyper-raising (Mongolian):
[Baatar$_i$ [$_{CP}$ öör$_i$-ACC [$_{TP}$ t_i buruud-san] gež] med-sen]
c. Nominalization (Mongolian):
[Baatar$_i$ [$_{DP}$ öör$_i$-GEN [$_{TP}$ buruud-san]-ig] meder-sen]

The preceding discussion demonstrates that reflexivity is obtained exclusively when bound items occupy a phase edge, assuming that DP and CP are phases. This applies equally when binding targets the edge of a phase nested within another phase edge. This phase-in-a-phase-edge (PIPE) binding is schematized in (62) and exemplified by (63–65), where RX marks reflexivity.

(62) PIPE binding: binder$_i$ … [$_{XP(phase)}$ [$_{YP(phase)}$ [Z]$_i$ Y] X …][34] (XP=CP or DP; YP=DP)



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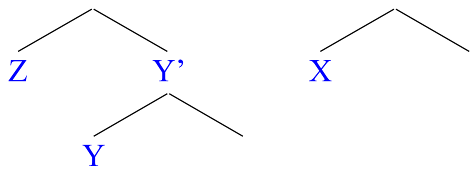
[33] See Bai et al. (2026) for detailed discussion of the Case puzzle: why $D_{[phi]}$ within Spec of the possessive DP structure is not genitive (*hisself*) but accusative (*himself*) on the surface.

[34] In principle, this nested PIPE structure is recursive; its empirical instantiation, however, is constrained, arguably by morphophonological factors.



(63) a. Baatar (öör-in) bagš-ø-aa buruud-san gež med-sen.
Baatar self-GEN teacher-ACC-RX go wrong-PST COMP know-PST
'Baatar realized that his (=Baatar's) teacher was wrong.'

b. Baatar$_i$ … [$_{CP}$ [$_{DP}$ öör$_i$ D(-in) [$_{NP}$ bagš-ø-aa]] [$_{TP}$ [$_{vP}$ …
(*öör* in Spec of DP in Spec of CP)

(64) a. Baatar (öör-in) bagš-in-aa buruud-san-ig med-sen.
Baatar self-GEN teacher-GEN-RX go wrong-PST-ACC know-PST
'Baatar realized that his (=Baatar's) teacher was wrong.' (Bao & Bai 2025: 265)

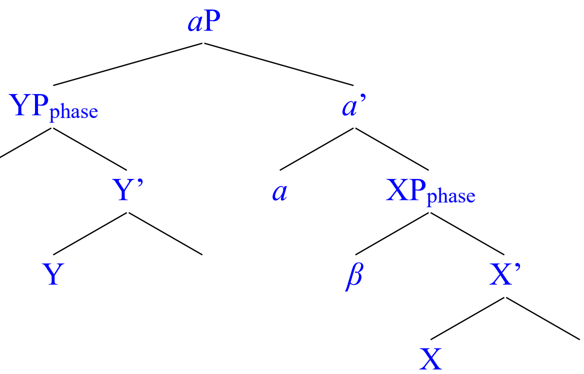
b. Baatar$_i$ … [$_{DP1}$ [$_{DP2}$ öör$_i$ D(-in) [$_{NP}$ bagš]] D(in)-aa [$_{TP}$ [$_{vP}$ …
(*öör* in Spec of DP2 in Spec of DP1)

(65) a. #Baatar öör-in bagš-in-aa nom-ig mart-san.[35]
Baatar self-GEN teacher-GEN-RX book-ACC forget-PST
'Baatar forgot his (=Baatar's) teacher's book.'

b. Baatar$_i$ … [$_{DP1}$ [$_{DP2}$ öör$_i$ D(-in) [$_{NP}$ bagš]] D(in)-aa [$_{NP}$ …
(*öör* in Spec of DP2 in Spec of DP1)

Importantly, a bound element is not necessarily located in a PIPE item, as it may reside entirely outside a leftmost (and simultaneously highest) phase (66).

(66) Non-PIPE binding: binder$_i$ … [$_{a\text{P}\neq\text{phase}}$ [$_{\text{YP=phase}}$ [Z]$_i$ Y] *a* [$_{\text{XP=phase}}$ X …]]



In (66), the binder binds into the edge of the YP phase, which is located outside the XP phase

[35] While the occurrence of the genitive anaphor renders the sentence less natural, it does not lead to ungrammaticality. The senrence is not fully accepted arguably due to the double occurrences of the genitive marker.

rather than at its edge. This is precisely what occurs in the ECM construction (67), where *himself*, projecting as a DP, resides outside the embedded vP phase, while *him* occupies the edge of that DP.

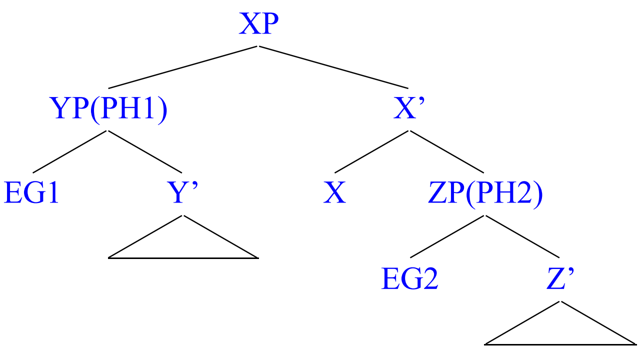
(67) a. Max expects himself to pass the exam.
b. Max$_i$ expects [$_{TP}$ [$_{DP}$ him$_i$ [$_{NP}$ self]] to [$_{vP}$ pass the exam]].

Taken together, (62) and (66) indicate that the bindee, Z, must reside either in the specifier of the highest phase (62) or in the specifier of a higher non-phasal projection c-commanded by the binder (66). In other words, reflexivity arises only when no phase intervenes between a binder and a bindee occupying specifier positions. Crucially, however, being the "highest" element is a sufficient but not a necessary condition for an item to be reflexively bound. Instead, being "leftmost" constitutes the true necessary condition; that is, regulating reflexivity strictly requires positioning the bindee at the edge of a leftmost phase, as illustrated below.

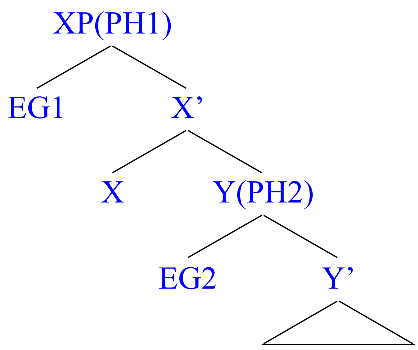
In (68a), the YP phase c-commands Z, the head of the ZP phase; therefore, EG1, the edge of YP, is higher than EG2, the edge of ZP. The structure in (68a) can thus represent the position of bound ECM subjects in English, where XP represents the embedded clause and YP represents the ECM subject. In (68b), the XP phase c-commands Y, the head of the YP phase; therefore, EG1, the edge of XP, is higher than EG2, the edge of YP. This configuration can represent the position of hyper-raised subjects (61b) and subjects of nominalized clauses (61c), where XP represents CP or DP, and EG1 represents the subject. However, (68b) and (68b) cannot represent the position of subjects with a nested PIPE structure (63–65). A nested PIPE structure is correctly represented in (69), in which EG1 is lower than DP, the edge of the CP phase, yet it, rather than DP, is reflexively bound. EG1 can be reflexively bound because it is located in the leftmost phase edge. This also holds in (68a) and (68b).

(68) EG1 is both leftmost and highest:

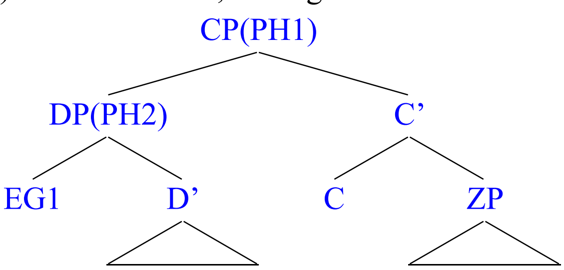
a.



b.



(69) EG1 is leftmost, not highest:



It then follows from the Mongolian data (PIPE binding) and the English data (binding of ECM subjects) that:

(70) Phase-edge Non-intervention Condition (PNC):[36]
Reflexivity arises only when there is no intervening phase leftmost edge between the binder and the bindee located in specifier positions.

It is thus reasonable to conclude that PNC serves as the true source of the locality required for reflexive binding. Consequently, Reuland's Agree-chain formation becomes unnecessary; the empirical patterns accommodated or left unresolved by the chain-based analysis can be fully accounted for under PNC and LEC.

3.4. Reflexivity as a continuum

A tension arising between coargumental reflexivity theory and the unified analysis proposed here concerns how SELF and SE anaphors license reflexivity under the analyses proposed in preceding subsections. Crucially, so-called "SE anaphors" are not expected to map onto the structures in (45–48), as they lack overt possessive marking. In resolving this tension, the crux of the matter is to note that the possessive structure diverges between SO and SP reflexivity, as evident in Mongolian and many other languages (especially those with body-anaphors and HIS-SELF type anaphors) in that it contains a minimal noun N for SO reflexivity (Bai et al. 2025), while it contains a full, content noun NP for SP reflexivity (Bai et al. 2026). Additionally, it tends to be more compact than for SO than SP reflexivity in terms of morphological exponent. In this sense, anaphors can be regarded as a condensed version of the possessive structure containing a metaphorical possessor in its specifier, which explains their status as an object argument for SO reflexivity, in addition to their inherent possessive structure. That is, there is blurring between SP and SO reflexivity to varying extents within or across languages. Specifically, the distinction between SP and SO reflexivity blurs where **some** anaphors can be structurally analyzed as possessive DPs while others cannot.

(71) Blurring between SO and SP reflexivity:

<table>
<tr><td rowspan="2">SO reflexivity</td><td>For decompositional object anaphors</td><td>Arises between subjects and objects</td></tr>
<tr><td>For non-decompositional object anaphors</td><td rowspan="2">Arises between subjects and possessors</td></tr>
<tr><td colspan="2">SP reflexivity</td></tr>
<tr><td colspan="2">SS reflexivity</td><td>Arises between subjects</td></tr>
</table>

It is thus evident that SS/SP and SO reflexivity together constitute a reflexivity continuum. In

[36] PNC is on par with the PIC (Chomsky 2001), serving as a condition complementary to the latter. Crucially, the chain-based analysis violates the PIC because it overlooks the phasal status of DPs; specifically, it argues that the presence of *self* (NP) inside *himself* (DP) is licensed by an external A-chain (Section 1). Under the possessive DP analysis proposed here, no syntactic chain intertwines with *self* within *himself*, thereby satisfying PIC. Furthermore, *him* represents the spell-out form of $D_{[phi]}$ at the DP edge, which likewise satisfies PIC. PNC also aligns with Bošković's (2016) formulation.

(i) [A]n anaphor can be bound outside of its own minimal phase XP only if it is located at the edge of the phase (the anaphor then does not really 'belong' to phase XP; rather, it belongs to a higher phase).
(Bošković 2016: 14)

this model, SS and SO reflexivity represent the two poles of the continuum, as shown in (72).

(72) Reflexive continuum (in terms of structural relation):
SS reflexivity → SP reflexivity → SO reflexivity

In terms of morphology, the reflexive form ranges from complex (possessive to compounding) to simplex (weak anaphor to clitic to affix) to zero along the continuum, as shown in (73).

(73) Reflexive continuum (in terms of reflexivizers):
Possessive anaphor (strong) → compounding anaphor (strong) → weak anaphor → clitic → voice affix → zero

It follows, then, that there are two dimensions to reflexive continuum: the continuum of reflexivity and the continuum of reflexivizer. Importantly, the blurring of the possessor-possessum relation gives SP reflexivity an intermediate status, effectively bridging the gap between SO and SS reflexivity.

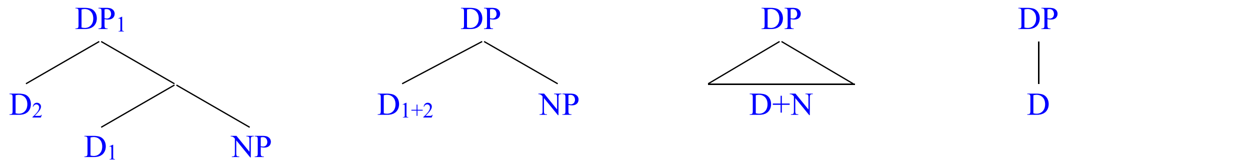
(74) The reflexive continuum:

(the continuum of reflexivity/coreference)
Subject
↓
Possessor
↓
Object (strong anaphor → weak anaphor → clitic → voice affix → null object) (the continuum of reflexivizer)

All this suggests that reflexivity is not a rigid category but rather a notion with a gradable scale. It is not a homogeneous category. Instead, it is a highly heterogeneous and multifaceted linguistic phenomenon. While it refers to a dependency where two elements share the same reference, how this is conditioned by syntactic (and pragmatic as well) constraints or principles and expressed/licensed by morphological forms varies greatly across languages. Thus, the continuum nature of reflexivity leads to the cross-linguistic variation of reflexivizers/reflexive licensers that range from anaphors to clitics, and to null forms. When language-specific factors such as the syntactic organization and the morphological inventory of either the nominal or verbal domain come into play, it is reasonable to say that licensing of reflexivity is not universal, contrary to what Reinhart & Reuland (1993: 662) and others have claimed. What matters instead is which configuration in (75a–75d) possesses a corresponding exponent that licenses coreference between two items in a given language. That is, a language may have lexicalized one or more of these structures (or specific nodes within a particular structure), whereas other languages, such as Khanty, Frisian, and Old English, may lack them entirely (Volkova & Reuland 2014). Crucially, not every exponent of these structures qualifies as an "anaphor" in the proper sense of the term. In many languages, a single form performs multiple functions; for instance, *himself* can surface as a bound variable, a logophor, or an intensifier. Consequently, one should not classify an expression as an anaphor simply because the existing literature labels it as such.

(75) Gradability and complexity of the structure of "anaphors" and reflexivizers:

a. b. c. d.



In short, "anaphors" can be defined and characterized in alternative ways; while they are mostly bound variables, they are not exclusively so. Sometimes, expressions of direct coreference or pragmatic covaluation that lack structural binding are termed "logophors", which share their surface form with bound variables (e.g., *The picture of himself in the newspaper upset John*). Traditionally, anaphors have been divided into SE and SELF types based on their morphosyntactic properties, with SE anaphors often being more structurally compact and freer from local binding than their SELF counterparts. However, the crucial regulator of reflexivity may not be this SE-SELF distinction at all. Instead, it is the possessive (or compounding) nature of a bound variable that plays the primary role in licensing reflexivity. If a bound variable lacks this internal possessive, it is more likely to escape strict locality constraints, even if it is still loosely referred to as an "anaphoric form" in the literature.

4. Conclusion

The crucial facts concerning Mongolian reflexive binding strategies are as follows. **First**, a uniform dedicated from RX (*-AA*) is used to mark local coreference between a subject and another item, which is the object of the same predicate, an adpossessor, or the subject of an embedded clause. **Second**, the following properties indicate that RX manifests Condition A of the Classical Binding Theory: It resists affixing to a nominative item, it is only attached to locally bound items, and it is undermined by disjoint reference. **Third**, subjects of embedded clauses and possessors within possessive DPs are invariably local to the (matrix subject). This locality constraint is satisfied by either nominalizing the embedded clauses or hyper-raising their subjects. It is naturally satisfied by locating possessors in the specifier of possessive DPs including anaphoric objects and regular possessives. Crucially, bound items in reflexive environments must be located at the edge of a leftmost DP or CP phase.

These reflexive properties reveal the following key insights regarding the nature of reflexivity. **First**, reflexivity includes both coargumental and non-coargumental coreference.[37] **Second**, reflexivity arises between a c-commanding specifier and the specifier of a leftmost c-commanded phase and is conditioned by binding as phi-Agreement targeting the specifier of the bound DP. Reflexivity relies on a phase-based locality condition. Consequently, a structural configuration sensitive to phasal locality plays a more critical role in regulating reflexivity than predicate argument structures or Agree-chain formation. **Third**, reflexivity may or may not be licensed. Where licensed, the licenser can be anaphors, clitics, or verbal affixes, which constitute a morphosyntactic reflexivizer continuum. Anaphoric forms do not inherently function as reflexive licensers. This explains why anaphors cross-linguistically fail to restrict

[37] Coargumental reflexivity may be regarded as the primary type of reflexivity but it definitely is not the only type.

anaphorically marked coreference to local dependencies, why certain languages (like English) lack possessive anaphors to mark subject-possessor coreference within a local domain, and why languages (like Khanty, Frisian, and Old English) do not overtly mark subject-object coreference. **Fourth**, coreference is in principle free and exists along gradient scales, from discourse to syntax, and from logophoricity to reflexivity. When subjected to structural principles such as PNC and LEC, it is regulated as reflexivity. Consequently, reflexivity is not a homogeneous category but rather a heterogeneous, multifaceted linguistic phenomenon. It emerges as a byproduct of syntactic principles constraining coreference, manifesting as a continuum where subject-object and subject-subject dependencies represent the two extremes. The bare-phrase status of D heads, which blurs the head-phrase boundary, allows them to project as specifiers of the possessive D head. This projects a blurred possessor-possessum relation that effectively bridges the two extremes of the continuum. Briefly stated, syntactic locality principles or constraints regulating coreference as reflexivity is universal, whereas specific licensing modes and the locality requirements of them are subject to parametric variation

In sum, Mongolian demonstrates the following properties in (76) regarding reflexivity, thereby resolving our earlier questions repeated in (77) and the Uniformity and Possessive Puzzles.

(76) a. Reflexivity configurations include subject-object, subject-possessor, and subject-subject relations; reflexive predicates are not universal.
b. Reflexivity occurs only when no phase (and hence phase edge) intervenes between the binder and the bindee.
c. Reflexivity is not universally licensed; it triggers but does not strictly require a licenser; conversely, anaphors may license reflexivity but do not enforce it.
d. Reflexivity is a continuum, where the blurring of the possessor-possessum relation and the external-argument nature of possessors bridge subject-object and subject-subject reflexivity.

(77) a. What is the mechanism that regulates and unifies coargumental and non-coargumental coreference as reflexivity?
b. Why is reflexive licensing is non-universal and why do licensing modes vary cross-linguistically among anaphors, clitics, and null forms?
c. What roles do locality and binding play in regulating and licensing reflexivity?

Given the empirical and theoretical difficulties with the chain-based analysis show that it cannot be entirely correct. These difficulties stem directly from its conceptually problematic foundational assumption that "*protection* by SELF" correlates a phase-internal structure with a phase-external syntactic operation (Agree-chain). In contrast, the proposed analysis, which underscores the correlation between the notion of a phase and the possessive nature of DPs, successfully captures various cross-linguistic facts that remain problematic for the chain-based framework.

Their core difference is that the proposed analysis claims that binding-regulated reflexivity must strictly obey syntactic principles like PNC, not only across categories such as DP, CP, and vP, but also across SO, SP, and SS coreference. This is not true of the chain-based analysis. Regarding why SP coreference is more likely to be unmarked than SO/SS coreference cross-linguistically, the proposed analysis posits that this variance stems from the definiteness-marking parameter. For example, the lack of possessive anaphors in English is attributed directly to the language's obligatory definiteness marking.

Specifically, the bare D head with phi-features in the specifier of the possessive D head (78) is obligatorily spelled out due to the definiteness-marking requirement, whereas the D head with the reflexivity feature need not be spelled out. In contrast, in other languages like Chinese (79a), along with Mongolian (79b), and, either or both of these heads can be spelled out in a coreferential environment.

(78) Derivation of possessive DPs (including anaphors):

a. b.

(79) Spell-out possibilities:

a. ta-ziji-de shu 'he-self's book'
ta-de shu 'he's book'
ziji-de shu 'self's book'

b. tüün-ne öör-in nom-oo 'he's self's book-RX'
tüün-ne nom-oo 'he's book-RX'
öör-in nom-oo 'self's book-RX'
nom-oo 'book-RX'

Notably, when there is no need to intensify the coreference relation or the possessor's identity, economy principles favor more compact forms, as argued by Bai et al. (2026). Note that while the definiteness-marking parameter crucially explains cross-linguistic variations in overt spell-out requirements, it does not dictate structural identity between anaphoric object DPs and regular possessive DPs as the chain-based analysis proposes.[38] Under the proposed analysis, these two DP types are structurally identical, ensuring a uniformity of licensers; their structural identity does not inherently correlate with the definiteness-marking parameter, but rather with the bare-phrase status of D heads, which blurs the head-phrase boundary. In contrast, the chain-based analysis treats them as fundamentally distinct by attributing the difference to this parameter, yet it fails to resolve both the Uniformity and Possessive Puzzles.

[38] Note that under the chain-based analysis, pronominal possessors spell out the D head of the entire DP structure, whereas under the proposed analysis, they spell out bare D heads with phi-features structurally projected as specifiers (§3.2). Thus, pronominal possessors are "barriers" to external operations under the chain-based analysis, whereas they are not under the proposed analysis.

Abbreviations

ABL: ablative case
ACC: accusative case
ASP: aspect morpheme
CS: causative
DAT: dative case
GEN: genitive case
INS: instrumental case
NEG: negation particle/suffix
NFIN: non-finite form
NOM: nominative case
PS: passive morpheme
POSS: possessive suffix
PST: past tense morpheme
RX: reflexive-possessive suffix
ø: lack of phonological content

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